

Hoigi Seo

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EDUCATION

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- | | |
|---|---|
| • Seoul National University
<i>B.E. in Electrical and Computer Engineering</i> | Seoul, Republic of Korea
<i>March 2018 - February 2024</i> |
| • Seoul National University
<i>Ph.D. Candidate in Electrical and Computer Engineering</i> | Seoul, Republic of Korea
<i>March 2024 - Present</i> |

RESEARCH INTEREST

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- **Multi-Modal Generative Models:** Multi-modal large language model, Text-to-image / video diffusion model
 - **Efficient Model Training & Inference:** On-Policy Distillation, Quantized model training, Diffusion model acceleration
 - **Efficient Agentic System:** Small Language Models (SLMs), Multi-Agent System, Agentic Swarm

PUBLICATIONS (SELECTED)

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- **Strong Helps Weak: Directional Cross-Modal Alignment Transfer in Multi-modal LLMs**
Hoigi Seo*, B H Lee*, M Kim*, D Mah, J Lee & S Y Chun. (*co-first authors)
Neural Information Processing Systems (**NeurIPS**), **Spotlight (Top 1.3%)**, 2026.
 - **Triadic Dynamics Aware Diffusion Posterior Sampling for Inverse Problems: Optimizing Guidance and Stochasticity Schedules**
J Bang*, D Moon*, **Hoigi Seo**, S Hong & S Y Chun. (*co-first authors)
International Conference on Machine Learning (**ICML**), 2026.
 - **Erasing Thousands of Concepts: Towards Scalable and Practical Concept Erasure for Text-to-Image Diffusion Models**
Hoigi Seo*, B H Lee*, J Cho, S Lim & S Y Chun. (*co-first authors)
Conference on Computer Vision and Pattern Recognition (**CVPR**), 2026.
 - **Training free, Perceptually Consistent Low-Resolution Previews with High-Resolution Image for Efficient Workflows of Diffusion Models**
W Jeong*, **Hoigi Seo*** & S Y Chun. (*co-first authors)
Conference on Computer Vision and Pattern Recognition (**CVPR**), 2026.
 - **Training-free Mixed-Resolution Latent Upsampling for Spatially Accelerated Diffusion Transformers**
W Jeong*, K Lee*, **Hoigi Seo** & S Y Chun. (*co-first authors)
Conference on Computer Vision and Pattern Recognition (**CVPR**) **Highlight (Top 3.4%)**, 2026.
 - **On Epistemic Uncertainty of Visual Tokens for Object Hallucinations in Large Vision-Language Models**
Hoigi Seo*, D U Kang*, H Cho, J Lee & S Y Chun. (*co-first authors)
Neural Information Processing Systems (**NeurIPS**), 2025.
 - **Skrr: Skip and Re-use Text Encoder Layers for Memory Efficient Text-to-Image Generation**
Hoigi Seo*, W Jeong*, J Seo & S Y Chun. (*co-first authors)
International Conference on Machine Learning (**ICML**), 2025.
 - **Efficient Personalization of Quantized Diffusion Model without Backpropagation**
Hoigi Seo*, W Jeong*, K Lee & S Y Chun. (*co-first authors)
Conference on Computer Vision and Pattern Recognition (**CVPR**), 2025.
 - **INTRA: Interaction Relationship-Aware Weakly Supervised Affordance Grounding**
J H Jang*, **Hoigi Seo*** & S Y Chun. (*co-first authors)
European Conference on Computer Vision (**ECCV**), 2024.
 - **BeyondScene: Higher-Resolution Human-Centric Scene Generation with Pretrained Diffusion**
G Kim*, H Kim*, **Hoigi Seo***, D U Kang* & S Y Chun. (*co-first authors)
European Conference on Computer Vision (**ECCV**), 2024.

PREPRINTS

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- **S³-OPSD: Sparse, Stable and Speedy On-Policy Self-Distillation**
Hoigi Seo, B H Lee, M Oh & S Y Chun.
ICLR Under review, 2027.
 - **DNA-RADIO: Towards an Agglomerative DNA Embedding Model**
Hoigi Seo, H Bae, K Lee & S Y Chun.
ICLR Under review, 2027.
 - **Speculative Encoding for Efficient Gigapixel Whole Slide Image Analysis**
Hoigi Seo, H Bae, B H Lee, J Cho, J Lee, Y Kim, S Y Jeon, J H Jang, H Kim & S Y Chun.
ICLR Under review, 2027.

- **HYPERION: Joint-Hierarchy Hyperbolic Fine-Tuning for Histopathology Slide Representations**
J H Jang, H Kim, S Y Jeon, Y Kim, J Cho, J Lee, **Hoigi Seo**, H Bae, B H Lee & S Y Chun.
ICLR Under review, 2027.
- **Recovering Diversity in Distilled Diffusion Models via Low-Rank Policy-Optimized Merging**
Hoigi Seo, W Han, J Lee & S Y Chun.
WACV Under review, 2027.
- **Geometrical Properties of Text Token Embeddings for Strong Semantic Binding in Text-to-Image Generation**
Hoigi Seo*, J Bang*, H Lee*, J Lee, B H Lee & S Y Chun. (*co-first authors)
arXiv preprint, 2025.
- **DITTO-NeRF: Diffusion-based Iterative Text To Omni-directional 3D Model**
Hoigi Seo*, H Kim*, G Kim* & S Y Chun. (*co-first authors)
arXiv preprint, 2023.

SKILLS SUMMARY

- **Programming Languages:** Python, C, C++, Bash
- **Deep Learning Frameworks:** PyTorch, TensorFlow
- **Distributed Training:** PyTorch DDP / FSDP, DeepSpeed, HuggingFace Accelerate
- **Inference & Serving:** vLLM, quantization (GPTQ / AWQ / bitsandbytes)
- **MLOps & Tooling:** Docker, Git, Linux, Weights & Biases, NVIDIA GPU cluster (A100 / H100)
- **Languages:** English (professional), Korean (native)

PATENTS

- **Speculative Encoding-Based Method and Apparatus for Whole Slide Image Analysis.**
S Y Chun, **Hoigi Seo**, H Bae, B H Lee, J Cho, & J Lee
Korean Patent, Filed. No. 10-2026-0107545, 2026.
- **Electronic Apparatus Including an Artificial Intelligence Model for Generating an Image and Operating Methods.**
S Y Chun, W Jeong, K Lee & **Hoigi Seo**
Korean Patent, Filed. No. 10-2026-0103476, 2026.
- **Electronic Apparatus for Generating Images and Operating Methods.**
S Y Chun, W Jeong & **Hoigi Seo**
Korean Patent, Filed. No. 10-2026-0106226, 2026.
- **Device and Method for Affordance Grounding.**
S Y Chun, J H Jang & **Hoigi Seo**.
Korean Patent, Filed. No. 10-2025-0029227, 2025.
- **Device and Method for Extracting High-Dimensional Gene Features.**
S Y Chun, **Hoigi Seo**, H Bae, D U Kang & H Kim.
Korean Patent, Filed. No. 10-2025-0014990, 2025.
- **Higher-Resolution Human-Centric Scene Generation With Pretrained Diffusion.**
S Y Chun, G Kim, H Kim, **Hoigi Seo** & D U Kang.
Korean Patent, Filed, No. 10-2024-0115484, 2024 / U.S. Patent Application, Filed, No. 19/019,060, 2025.

HONORS AND AWARDS

- Compute Transparency Champion Award (CVPR26) | June 2026
- Excellent Research Talent Fellowship (Korea Research Foundation) | September 2025
- Brain Korea 21 Scholarship (Korea Research Foundation) | March 2024 - Present
- Academic Excellence Scholarship (Seoul National University, ECE) | September 2023

RESEARCH PROJECT EXPERIENCE

● Uncertainty-aware World Model and VLA Model <i>Korea Electronics Technology Institute (KETI), Republic of Korea</i> ○ Building uncertainty-aware world model and VLA model that can estimate current state’s uncertainty and account for the uncertainty for the next action and prediction.	Project Lead <i>July 2026 - Present</i>
● Megabase-scale DNA foundation model <i>NVIDIA Academic Grant (32K A100 hours)</i> ○ Building foundation models for Megabase-scale DNA foundation model for structural variant detection.	Project Lead <i>April 2026 - Present</i>
● ecDNA Detection <i>Ministry of Science and ICT, Republic of Korea</i> ○ Extra-chromosomal DNA (ecDNA) detection and associated gene discovery from FISH images and gene profiles.	Project Lead <i>September 2023 - Present</i>
● AI for Human-Machine Teaming <i>United States Air Force & IITP, Republic of Korea</i>	Project Participant <i>March 2025 - Present</i>

- Developing human-robot interaction (HRI) and decision-making that can enable robots to collaborate naturally with humans with hazy oracle.
- **Efficient and Performative Pathology Foundation Model** Project Participant
National IT Industry Promotion Agency (NIPA), Republic of Korea *January 2026 - June 2026*
 - Building efficient digital pathology inference method and train 7B-parameter histopathology multi-modal large language model.
- **Efficient Personalization** Project Participant
Samsung Mobile Experience (MX) *March 2024 - May 2025*
 - Memory and time efficient text-to-image synthesis and personalization for on-device deployment.
- **Industrial Defect Segmentation** Project Participant
Samsung Device Solution (DS) *March 2024 - March 2025*
 - Development of industrial defect segmentation model that captures small defects in production.

OTHER PUBLICATIONS

- **Tunable Porous Collagen Hydrogels as a Physiologically Relevant Platform for Extrachromosomal DNA-Associated Colorectal Cancer Research**
S Jo, J Shon, S An, Y Nam, D Choi, S Lee, **Hoigi Seo**, S Y Chun & H Kim.
Theranostics (IF: 13.3), 2025.
- **ecSegCls: Deep learning-based method for detecting extrachromosomal DNAs in both interphase and metaphase cancer cells**
Hoigi Seo, Y Nam, H Bae, D U Kang, J Lee, H Kim & S Y Chun.
American Association for Cancer Research (**AACR**), 2026.
- **Harmonization for a black-box deep learning model**
M Kim, H Jeong, **Hoigi Seo**, W Jeong, S Y Chun & J Lee.
International Society for Magnetic Resonance in Medicine (**ISMRM**) (**Oral, Summa Cum Laude, AMPC select**), 2025
- **Mitigating Hallucinations in Large Vision-Language Models via Norm-Guided Visual Filtering**
H Cho, **Hoigi Seo**, D U Kang, J Lee & S Y Chun.
Conference on Electronics, Semiconductor and AI, 2025.
- **Towards Personalization of Generative Models Via Zeroth-Order Optimization**
W Jeong, **Hoigi Seo** & S Y Chun.
Korea Signal Processing Conference, 2024.

ACADEMIC SERVICE

- **CVPR, ECCV, NeurIPS, WACV, AAAI** Reviewer
Served as a reviewer for top-tier AI conferences. *2026 - Present*

MEDIA

- **Erasing 2,072 concepts on pre-trained diffusion model** Interviewee
Popular Science Korea *June 2026*
 - Discussed ETC, our CVPR 2026 method that erases 2,072 concepts (949 celebrities, 693 art styles, 430 characters) from a pre-trained diffusion model without manual tuning, and its implications for safety- and copyright-driven AI control.

INVITED TALK

- **On Efficient Diffusion Model Workflow** Invited Speaker
Samsung MX Seminar *April 2026*
 - Presented a training-free technique that generates low-resolution previews perceptually consistent with the final high-resolution image, enabling early candidate selection and up to 3× faster diffusion model workflows.
- **On Epistemic Uncertainty of Vision Encoder of MLLMs** Invited Speaker
SNU Medical Imaging AI Symposium *January 2026*
 - Presented an analysis of epistemic uncertainty in vision encoders of multimodal large language models, showing its connection to object hallucinations and introducing a vision-encoder-only approach that identifies and suppresses uncertain visual tokens to improve hallucination robustness.

OTHER EXPERIENCE

- **Military Service** Sergeant
Served as a Situation Room Soldier in the 3rd Guard Battalion. *May 2020 - November 2021*